UNIVERSITY OF NOTTINGHAM · EEEE4077 MECHATRONICS PROJECT

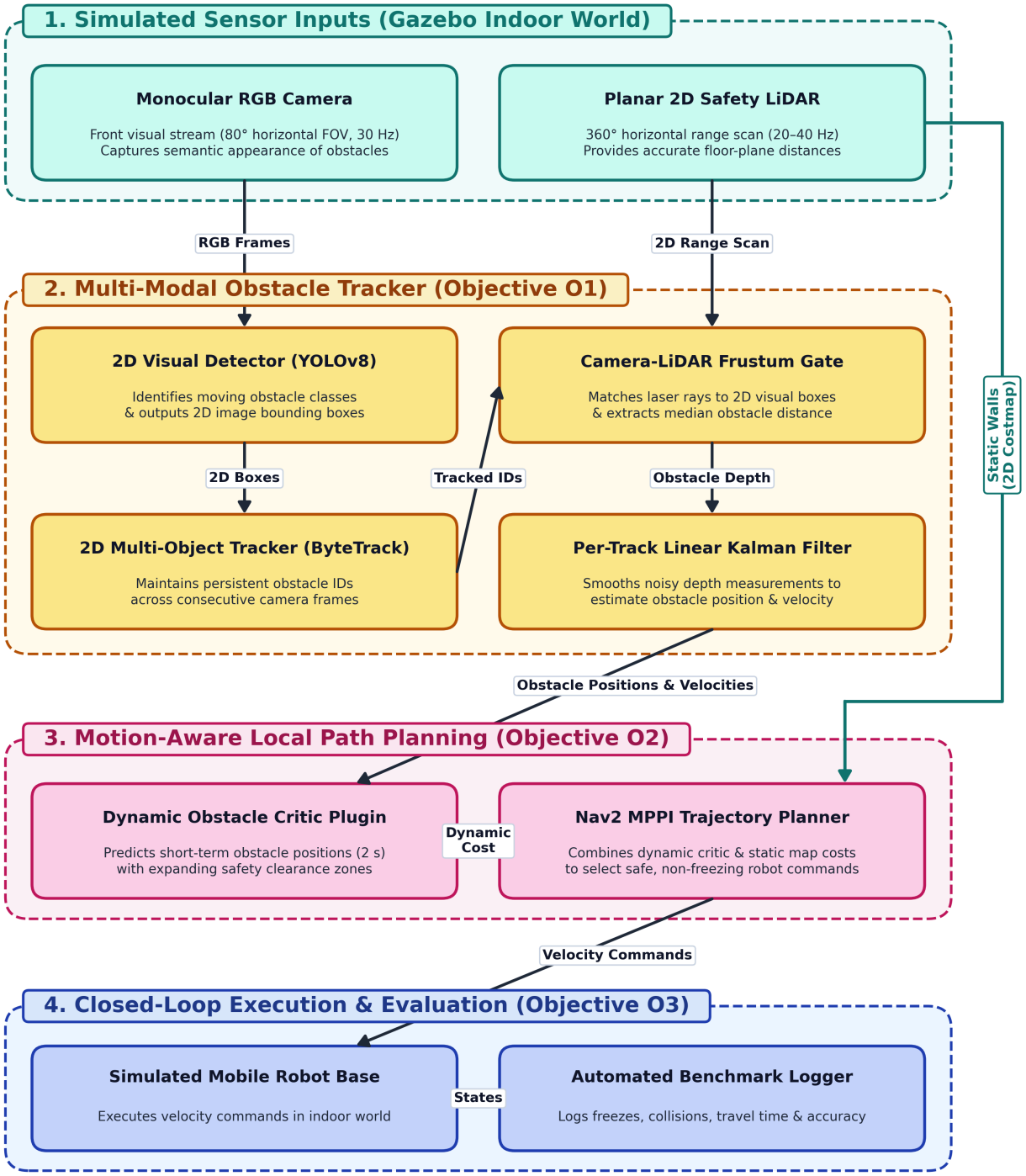
Vision-Augmented Mobile Robot Navigation in Dynamic Environments

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Established Space → Freezing Problem: Commercial AMRs rely on 2D safety LiDAR + static costmaps for walls. Because planar scans treat walking people as fixed obstacles, robots freeze or spin in shared aisles.

Vision-LiDAR Fusion (O1): Pairs existing 2D LiDAR with a monocular camera (YOLOv8 + ByteTrack + frustum gating) and Kalman filter for real-time obstacle velocities.

Predictive MPPI & Methods (O2–O3): Custom C++ critic projects motion 2.0s ahead to steer smoothly behind pedestrians on CPU—built and benchmarked via our 6-step workflow next.



6-STEP LIFECYCLE, FALLBACK & ABLATION PLAN

Methods and Approach

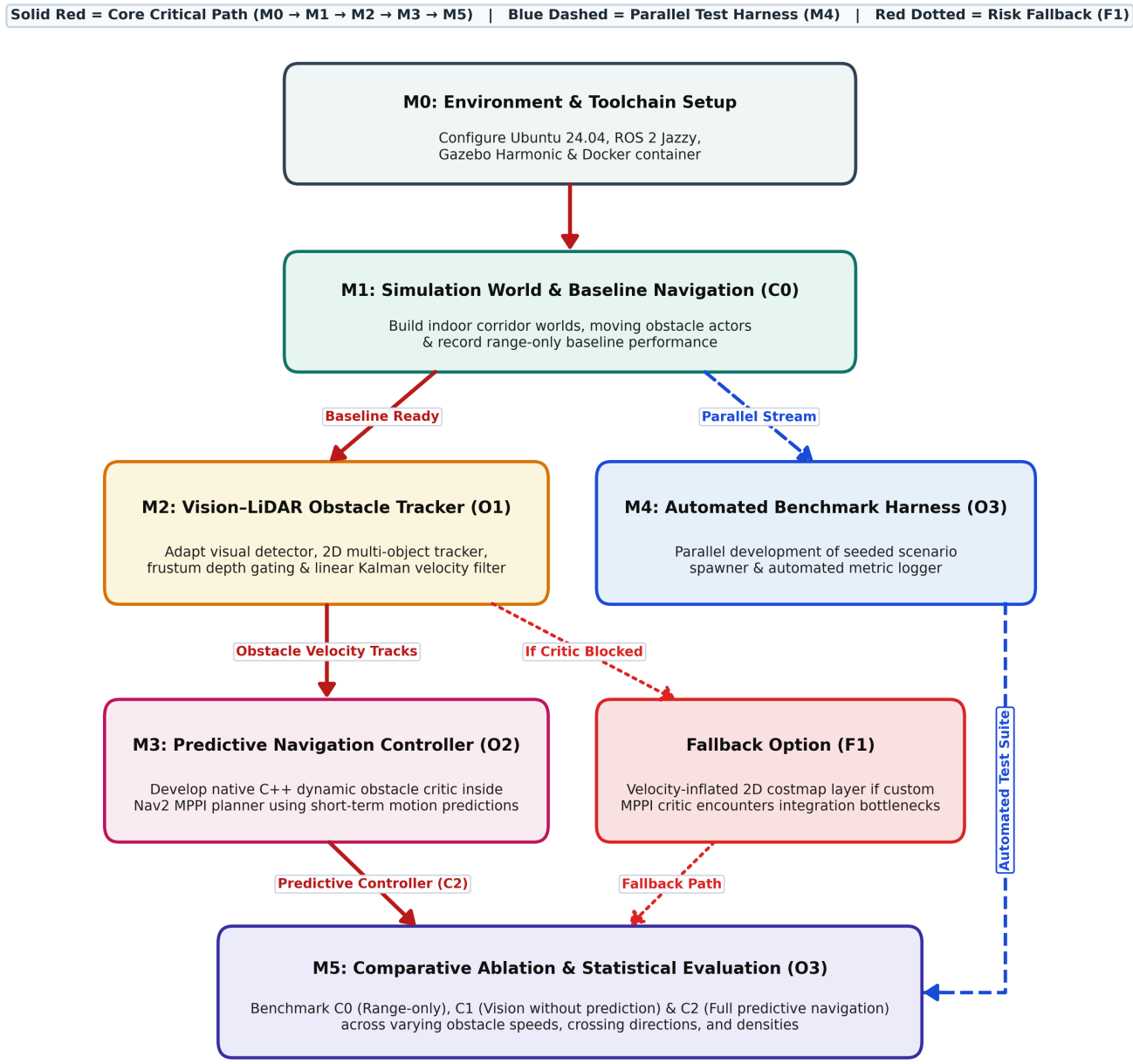
Critical Path (Solid Red): M0 (Docker) → M1 (Sim & C0) → M2 (Tracker O1) → M3 (MPPI Critic O2) → M5 (Statistical Analysis).

Parallel Test Harness (Blue): M4 automated runner (W11–20) executes Gazebo resets and rosbag2 logging for 450 unattended trials.

Fallback Architecture (F1, Dotted Red): If C++ MPPI critic hits bottlenecks, velocity bubbles are injected directly into standard 2D costmaps.

EXPECTED ABLATION EVOLUTION (C0 → C1 → C2)

ID	Parameters & Stack	Expected Navigation	Target Metrics & Limits
C0 Baseline	2D LiDAR (20 Hz) + static costmap; 2000 rollouts (τ = 0 s)	Frequent freezes (Δt ≥ 1.5 s, v < 0.05 m/s) & stop–go delays	0 collisions (via halts); 0 velocity tracking
C1 Vision	+ RGB (30 FPS, 80° FOV), YOLOv8 + ByteTrack + frustum (v = 0)	Cuts leg-split false stops; reactive yields only (no lookahead)	≥95% scan–ID assoc.; τ = 0 s clearance
C2 Proposed	+ 4-state Kalman CV + C++ critic (τ = 2.0 s, 40 steps, <0.5 ms)	**0 freezes, 15–25% faster** transit (≤5 s wait), smooth wake steering	**RMSE < 0.15 m/s**, d_{min} ≥ 0.35 m, TTC ≥ 1.2 s, **≥20 Hz** (450 runs, p < 0.01)



QUANTITATIVE RISK REGISTER · PRE- VS POST-MITIGATION RATINGS

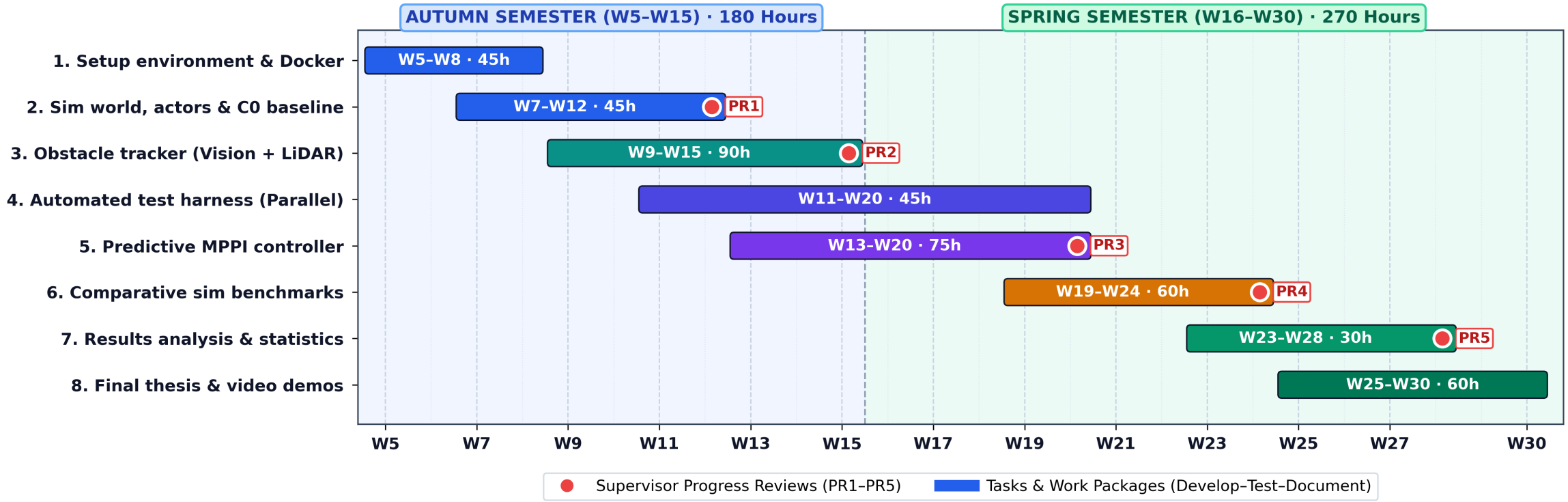
Risk Management and Mitigation

ID	Technical Risk Description	Timing	Pre-Score	Mitigation Actions & Fallback Strategy	Post-Score	Shift
R1	**C++ MPPI critic plugin integration fault or CPU budget overrun** (loop < 20 Hz)	W13–20	**20 (L4, I5)**	• Early minimal C++ prototype in Week 13. • Bounded 2.0s horizon + vectorised rollout scoring. • **Fallback F1:** Velocity bubbles in 2D costmap.	**6 (L2, I3)**	**-70%** (Low)
R2	**Camera–LiDAR projection error or laser ray bleeding on background walls**	W9–15	**12 (L3, I4)**	• Exact sensor transforms from Gazebo SDF. • Median depth filtering inside 2D frustum. • Kalman chi-square innovation outlier rejection.	**4 (L1, I4)**	**-67%** (Low)
R3	**Simulation instability or process crash during 450-trial batch evaluation**	W19–24	**12 (L3, I4)**	• Pinned Docker container for ROS 2 & Gazebo. • Headless Gazebo execution (cuts RAM by 60%). • Automated Python watchdog reset wrappers.	**4 (L1, I4)**	**-67%** (Low)
R4	**Visual occlusion between crossing obstacles causes track ID switching**	W11–15 & W21–24	**9 (L3, I3)**	• ByteTrack two-stage association matches low-confidence occluded boxes. • Kalman velocity coasting up to 1.0s.	**3 (L1, I3)**	**-67%** (Low)

Governance Takeaway: *All 4 technical risks mitigated to "Low" residual rating. Pure simulation scope (0% hardware) eliminates all physical Health & Safety hazards.*

PROJECT TIMELINE · 26 WEEKS · 450 PLANNED HOURS · 5 SUPERVISORY REVIEWS

Planned Work Schedule (Weeks 5–30)



Workload: 450 Hours Total
Autumn (W5–15): 180h (16.4 h/wk) · Spring (W16–30): 270h (18.0 h/wk)

5 Progress Reviews (PR1–PR5):
PR1 (W11): C0 Baseline · PR2 (W14): Tracker · PR3 (W19): MPPI Critic · PR4 (W23): Batch Runs · PR5 (W27): Thesis Draft

Final Safety Buffer:
Tests finish by W28; final 2 weeks reserved for thesis polishing